MIL-M-12624C(EL)
11 February 1972
Superseding
MIL-M-12624B(EL)
13 February 1970

MILITARY SPECIFICATION
MICROPHONE, RIBBON
(Microphone, T-23( ))

1. SCOPE

1.1 This specification covers one type of equipment designated as Microphone T-23( ) for detecting, amplifying, and transmitting sound of artillery fire and shell detonations. The microphone is part of Sound Ranging Set GR-8( ). (See 6.1 and 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture-and-Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	Varnish, Moisture-and-Fungus Resistant (For the Treatment of Communications, Electronics, and Associated Electrical Equipment)
MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-S-10956	Sound Ranging Set GR-8 Packaging and Packing of
MIL-P-11268	Parts Materials and Processes Used in Electronic Communication Equipment

FSC 5895

Source: https://assist.dla.mil -- Downloaded: 20261O-17T23:06Z
Check the source to verify that this is the current version before use.

MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	Wired equipment, classification of visual and mechanical defects.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.

DRAWINGS

ELECTRONICS COMMAND

SC-A-13474	Microphone T-23( ) Frequency Response Curve.
SC-DL-30620	Microphone T-23( ).
SC-A-46439	List of Accessories for Package Tester.
SM-C-336059	Wire Grid Assembly.
SC-A-678013	Burn-in/Debugging Procedure.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First articles.- The contractor shall furnish first articles for approval as required by the invitation for bids and contract. (See 4.3 and 6.2(c))

3.2 Construction.- The equipment shall be constructed in accordance with Drawing and Data List SC-DL-30620.

3.2.1 Standard microphone.- A microphone to be used as a standard during the tests of section 4 will be loaned to the contractor for the duration of the contract. (See 6.7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

3.3 Parts, materials, and processes; general.- In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Performance requirements.-

3.4.1 Dielectric strength.- The insulation of each completely assembled Microphone T-23( ) shall be capable of withstanding 500 volts DC and shall have a resistance not less than 10 megohms. After 5-second application of the voltage, there shall be no visible deterioration and the equipment shall meet specified performance requirements as determined by the test of 4.7.1.

3.4.2 Microphone element.- The microphone element shall be constructed in accordance with the drawing SM-C-336059 and meet the electrical requirements therein. (See 4.7.2)

3.4.3 Frequency response (with 25 Hz acoustic plug).- The microphone, with 25 Hz acoustic plug installed, shall have a frequency response over the range of 5 to 40 Hz such as to meet the test of 4.7.3.

3.4.4 Linearity.- The microphone, with both 25 Hz and 60 Hz acoustic plugs installed (not simultaneously), shall meet the tests of 4.7.4.

3.4.5 Plate current.- The plate current of the electron tube in the microphone shall decrease when positive sound pressure is impressed on the microphone. (See 4.7.5)

3.4.6 Amplifier phase.- The phase of the amplifier shall be such as to meet the test of 4.7.6.

3.5 Finish, protective.- Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc, and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3)

3.6 Marking.-

3.6.1 General.- Marking shall conform to MIL-M-13231. (See 4.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

3.6.2 Visibility.- Wherever practicalbe, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers. - Each microphone shall be serial numbered.

3.7 Tropicalization.-

3.7.1 Overall treatment (moisture and fungus proofing treatment (MFP).- Moisture and fungus resistant coating of varnish conforming to MIL-V-173 shall be applied in accordance with MIL-T-152. The moisture-fungus resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, and solder fluxes. When tape is used on parts of equipment that are masked against treatments, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

3.8 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.8)

3.9 Burn-in(debugging).- The microphone shall be capable of meeting the requirements of this specification after being subjected to the burn-in (debugging) cycle of Drawing SC-A-678013. (See 4.9)

3.10 Service conditions.- After a warmup period of 10 minutes, the microphone shall meet the performance requirements of this specification when subjected to the following conditions:

3.10.1 Temperature.-

3.10.1.1 Low temperature.- During testing as specified in 4.10.1, the microphone shall meet the performance requirement of 3.4.4.

3.10.1.2 High temperature.- During testing as specified in 4.10.2, the microphone shall meet the performance requirement of 3.4.4.

3.10.2 Humidity (moisture-resistance).- During testing as specified in 4.11, the microphone shall meet full specification performance for the following measurements:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

Paragraph	Measurement
3.4.3	Frequency response
3.4.4	Linearity

3.10.3 Bounce (vibration) and shock.- The microphone shall meet the requirements of table I.

Table I Bounce (Vibration) and Shock

Requirement	Insp Para	Performance after test
Bounce(vibration)	4.12	Full specification performance Minor mechanical damage is permissible.
Shock; ballistic	4.13	Full specification performance Minor mechanical damage is permissible.

3.11 Tools and running spare parts.- Tools and running spare parts shall be furnished as specified in the invitation for bids and contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2(d).)

3.12 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.

Cleaning prior to welding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

(b) In this specification:

Paragraph 3.2 Construction
Paragraph 3.6 Marking
Paragraph 3.8 Interchangeability
Paragraph 3.4.2 Microphone element

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance.- The contractor shall provide and maintain a means of determining product conformance in accordance with the requirements specified herein or elsewhere in the contract. The Government, at its option, may perform any evaluation deemed necessary to assure the adequacy of the means employed and the effectiveness of the contractor's quality assurance methods and systems.

4.1.2 Government verification.- All quality assurance operations required of the contractor shall be subject to Government verification at scheduled or unscheduled intervals. Verification will consist of the following:

a. Surveillance of the contractor's operations to determine that practices, methods and procedures of the contractor's quality assurance system are being applied and are, in fact, accomplishing the objectives of total compliance of the product with the requirements of this specification and the contract.

b. Government product inspections to determine compliance of the product with the specification and contract requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

4.1.3 Accomodation and assistance. The Government quality assurance representative shall have the right of access to any area of the contractor's or his subcontractor's premises where any part of the work is being performed. The Government quality assurance representative shall be afforded unrestricted opportunity to verify conformance of the product with specification requirements. The contractor shall make his inspection equipment and records available for use by the Government quality assurance representative for verification purposes. The contractor's personnel shall be made available for operation of such inspection equipment as required.

4.2 Classification of inspections.- Inspections shall be classified as follows:

(a) First article inspection. (Does not include preparation for delivery). (See 4.3)

(b) Inspections covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.6)

(2) Quality conformance inspection of preparation for delivery. (See 4.15)

4.3 First article inspection.- This inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for group A and group C (see tables IV and V respectively). First article inspection shall be performed in the order shown in table II according to the number of samples specified in the contract or order.

Table II Order of Environmental Testing

Inspection (Note 1)	Number of Units				
	Four		Nine		
	2 units	2 units	3 units	3 units	3 units
Non-damaging (Note 2)					
Low temperature	1		1		
High temperature	2		2		
Potentially damaging					
Humidity (moisture resistance)	3			1	
Bounce (vibration)		1			1
Shock; ballistic		2			2

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:06Z
Check the source to verify that this is the current version before use.

Note 1: Other inspections may precede, follow, or be interspersed between the non-damaging tests.

Note 2: The order of non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the bounce (vibration) test shall be performed on the same units used for the shock test.

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

Item	Where required
Finish	3.5
Marking	3.6

4.5 Equipment Verification Review.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings cited for construction. (see 3.2) The EVR shall consist of the following:

4.5.1 An audit to establish that "as-built" equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

4.5.2 An audit to establish that "as-built" configuration of unit, assembly, module, or part meet the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.5.3 The contractor shall compile and EVR report(s) containing the following information:

(a) Identification of unit, assembly, module, or part and details proving compliance with 4.5.1 and 4.5.2 above.

(b) Discrepancies noted.

(c) Corrective action taken.

4.5.4 The contractor shall make available to the Government copies of his EVR report(s) no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

(a) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(b) As required, responsible personnel from each functional department available for discussions in their respective areas.

(c) Adequate administrative support for the EVR.

4.6 Quality conformance inspection of equipment before preparation for delivery.- Every equipment shall be subjected to burn-in (debugging) (see 3.9) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.6.1 through 4.6.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of burn-in (debugging) shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records as an element of Government verification (see 4.1.2).

4.6.1 Group A inspection.- Equipment shall be assembled into discrete lots for quality conformance inspection, utilizing the criteria of MIL-STD-105 for determination of lot composition. Unless otherwise specified, the lot size shall be determined from table III, as related to production rates necessary to meet delivery schedules of the contract. The lot size shall be adjusted throughout the life of the contract as necessary to provide for changes in contract delivery schedules and production rates. Each unit of each lot of equipment shall be inspected for conformance to all the examinations and tests required in table IV. The quality conformance of each lot shall then be subjected to an audit (see 6.9), utilizing the procedures of MIL-STD-105, under the general inspection levels and the Acceptable Quality Level (AQL) indicated in table IV. Group A inspection shall be performed in any order satisfactory to the Government, except that the linearity test shall be last.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

Table III Lot Formation

Production Rate	Size of Lot
500 or more per month	One week's production
51 to 499 per month	Two week's production
0 to 50 per month	One month's production

Table IV - Group A Inspection

Inspection or Test	Reqt Para	Insp Para	AQL	
			Major	Minor
Visual and mechanical	3.12	4.14	1.5dphu*	6.5dphu
Electrical				
Dielectric strength	3.4.1	4.7.1		
Microphone Element	3.4.2	4.7.2		
Frequency response (with 25 cycle plug)	3.4.3	4.7.3	1.5dphu for group	
Plate current	3.4.5	4.7.5		
Amplifier phase	3.4.6	4.7.6		
Linearity	3.4.4	4.7.4		
All electrical defects are considered major				
*Defects per hundred units				

4.6.2 Group B inspection.- Not required.

4.6.3 Group C inspection.- This inspection shall consist of the tests specified in table V and shall be performed on units of product that have been subjected to and passed Group A inspection requirements. Sample units shall be selected in accordance with paragraph 4.6.3.1.

Table V Group C Inspection

Inspection (Note 1)	Reqt Para	Insp Para
Group 1		
Low temperature	3.10.1.1	4.10.1
High temperature	3.10.1.2	4.10.2
Group 2		
Humidity (moisture resistance)	3.10.2	4.11
Bounce (vibration)	3.10.3	4.12
Shock; ballistic	3.10.3	4.13

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:06Z
Check the source to verify that this is the current version before use.

Inspection (Note 1)	Reqt Para	Insp Para
Group 3 Equipment verification review (Note 2)	3.2	4.5
Note 1: The order of inspection shall be in accordance with table II. Note 2: Equipment verification review shall only be performed on the first production lot.		

4.6.3.1 Sampling for group C inspection.- Two sample units shall be selected at random for each group of inspections in table V from every 160 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot (see 4.6.3).

4.6.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspections. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.6.3.3 Noncompliance.- The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the processes and materials, as applicable, and on all units of product which are manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded through the Government procurement quality assurance representative to the responsible technical activity designated in the contract. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the technical activity) and group A inspection may be reinstituted. However, final acceptance and shipment will be withheld until group C reinspection results have shown that the corrective action was effective and approved by the technical activity. (See "Delivery and Conditional Acceptance of Contract Items Subject to Group C Inspection" provisions elsewhere in the contract.)

4.6.4 Reinspection of conforming group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group C inspection, may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:06Z
Check the source to verify that this is the current version before use.

4.7 Performance tests.-

4.7.1 Dielectric test.- Short the microphone input terminals to each other, apply 500 volts DC between this short and the microphone ground for a period of five seconds. At the end of the five seconds the insulation resistance shall not be less than ten megohms. A megger can be used for this test. (See 3.4.1)

4.7.2 Microphone element.- The microphone element shall be tested by compliance with 3.4.2. The resistance of the microphone wound grids shall be measured in a bridge circuit in which the wound grids under test can be alternately switched in and out of the bridge circuit for comparison with each other.

4.7.3 Frequency response (with 25 Hz acoustic plug).- The microphone shall be tested for frequency response as specified below. The response shall not be less than 85% of the value shown on the response curve of the drawing. (See 3.4.3)

(a) Measure the voltage, under load, of the microphone internal battery. It shall be 9 volts.

(b) Place the microphone in the test chest (see 6.8).

(c) Connect the microphone in the circuit as shown in Drawing SC-A-13474.

(d) Operate the piston-phone in the test chest at frequencies covering the range of 5 to 40 Hz. Each frequency shall be determined accurately within an accuracy of ± 1% by measuring the flywheel velocity with a stroboscope. (Other approved means of measuring the piston frequency may be used). Measurement of the microphone output voltage shall be made at 5 Hz intervals of frequency; a smooth curve of frequency-response shall result.

4.7.4 Linearity.- Microphone T-23( ) shall be tested as described herein. At all frequencies between 5 and 40 Hz with 25 Hz acoustic plug installed and at all frequencies between 15 and 120 Hz with 60 Hz acoustic plug installed. Any variation in response between the standard (see 3.2.1) and the test microphone shall be within 2 db over the frequency ranges. At all frequencies in the specified ranges, the waveform of the test microphone output shall meet the test described in 4.7.4 (f)(2). The following test procedure shall be used:

(a) The supply to both microphones (standard and test) shall be 9 volts.

Source: https://assist.dla.mil -- Downloaded: 202640-17T23:06Z
Check the source to verify that this is the current version before use.

(b) Mount the standard microphone and the test microphone in the test chest. (see 6.8)

(c) Apply 90 volts DC in series with 5100 ohms resistance to the terminals of each microphone.

(d) Connect the terminals of both microphones, through a suitable switch, to a cathode-ray oscilloscope with constant response over a frequency range of 5 to 120 Hz. The connection shall be such that by operating the switch, the outputs of the standard microphone and the test microphone will appear alternately as vertical traces on the oscilloscope screen.

(e) Operate the piston-phone in the chest set at the following frequencies:
With 25 Hz plug: 5, 10, 15, 20, 25, 30, 35, and 40 Hz.
With 60 Hz plug: 15, 30, 40, 50, 60, 70, 80, 90, 100 and 120 Hz.

(f) At each frequency operate the switch to view alternately, on the oscilloscope screen, the output of the standard and test microphones. The horizontal sweep of the oscilloscope shall not be operating.

(1) Record theheight of the traces for the two microphones. The height of the trace shall be used as a measure of the response voltage.

(2) Observe the trace of the tested microphone for nonuniform variation in intensity along the line; such variation indicates unsatisfactory wave form of the microphone output, and shall be verified by operating the horizontal sweep of the oscilloscope. A satisfactory microphone shall produce a trace such that there is no visible difference between the waveforms of the microphone under test and the standard one.

4.7.4.1 Check of standard microphone.- On each day when production microphones are tested in accordance with 4.7.4 the standard microphone (see 3.2.1) shall be first checked to meet the test of 4.7.3. A similar check of the standard microphone shall be made during the day's production testing, whenever required by the Government inspector.

4.7.5 Plate current test.- With the microphone arranged as for the linearity test of 4.7.4 wave a hand across the microphone filter and check the resulting signal on the oscilloscope for conformance with the requirement of 3.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

4.7.6 Amplifier phase test.- The relative phase of each amplifier shall be compared with the standard amplifier. Deviations in phase shall not cause more than 0.001 second in time deviation between the response of the amplifier under test and the standard amplifier over the frequency range 15 to 200 Hz. Measurements shall be taken with an input signal of approximately 0.001 volt across the hot-wire unit. The output impedance of the generator used shall be 1000 ohms. The output signal of the standard microphone shall be connected to the input terminals of one channel of an oscilloscope that has the output of the microphone under test connected to the input terminals for the other deflection circuit of the same oscilloscope. The oscilloscope must not introduce phase shift over the frequency range tested. A direct coupled oscilloscope with both channels coupled to the microphones through 0.5 mfd coupling capacitors will meet the requirements. The Lissajou figure obtained on the oscilloscope shall have a horizontal deflection of 20 divisions of the scope screen. The vertical deflection shall be approximately 20 divisions (actual deflection is not critical). The horizontal spread at the center of the sweep in divisions shall not exceed the following values:

Frequency(Hz)	Spread (divisions)
15	1.9
35	4.4
85	10.2
200	19.0

A dummy hot-wire unit may be substituted in order to prevent the pick-up of acoustical signals. The dummy hot-wire unit should be two 50 ohm ± 1%, 1 watt resistors. (see 3.4.6)

4.8 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasterners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

Source: https://assist.dla.mil -- Downloaded: 202640-17T23:06Z
Check the source to verify that this is the currentversion before use.

(e) Size and form of special threads.

4.9 Burn-in(debugging).- All units submitted for acceptance shall be subjected to burn-in(debugging) in accordance with drawing SC-A-678013. (See 3.9)

4.10 Temperature tests.-

4.10.1 Low temperature.- The microphone shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. Prior to testing the microphone shall meet the performance requirement of 3.4.4. The storage temperature shall be -34.4°C(-30°F) and the operating temperature shall be -31.7°C(-25°F). The storage temperature shall be maintained for 24 hours. The rate of temperature change shall not exceed 11.1°C(20°F) per hour. Sources of noise shall be minimized during the test period.

4.10.2 High temperature.- The microphone shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. Prior to testing, the microphone shall meet the performance requirement of 3.4.4. The operating temperature shall be 65.6°C(150°F) and the storage temperature shall be 71.1°C(160°F). Sources of noise shall be minimized during the test period.

4.11 Humidity (moisture-resistance) test.- The microphone shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing the microphone shall meet the performance requirements as specified in 3.10.2. Measurement shall be made during the period at the end of the 2nd and 4th cycles. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. Sources of noise shall be minimized during the test period.

4.12 Bounce (vibration) test.- The microphone shall be tested in accordance with Method 514, Procedure XI, of MIL-STD-810. Prior to testing, the microphone shall meet full specification performance. Package tester accessories shall be selected from those listed on Drawing SC-A-46439. (See 3.10.3)

4.13 Shock test; ballistic.- The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" as shown in MIL-S-901. The equipment shall be secured in its normal operating position to the steel test plate. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the

Source: https://assist.dla.mil -- Downloaded: 202640-17T23:06Z
Check the source to verify that this is the currentversion before use.

test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. Prior to testing, the microphone shall meet full specification performance. (See 3.10.3)

4.14 Visual and mechanical inspection.- The equipment shall be examined for the defects listed in MIL-STD-252 and Table VI.

Table VI - Classification of visual and mechanical defects.-

Classification	Defects
Major	Interchangeability of parts. Improper fit of cover. Improper fit of acoustical plugs. Batteries loose in holders. Electron tube loose in socket. Loose binding posts. Loose plug connection. Improper fit of hot-wire element. Improper fit of lightning arresters.
Minor	Loose fasteners. Fasteners difficult to operate. Untidy wiring.

4.15 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, packaging, and marking.- Preservation, packing, packaging and marking shall be in accordance with MIL-S-10956(EL).

6. NOTES

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

6.1 Intended use.- Microphone T-23( ) is intended for use with other units of Sound Ranging Set GR-8 in the detection, amplifying, and transmitting of signals to the recorder.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Level A, level B, or level C).

(c) Number of first articles to be submitted for approval (see 3.1)

(d) Quantity of tools and running spare parts required. (see 3.11)

(e) Submission of the statement of treatment before first article approval. This statement should be submitted to the contracting officer. (See 3.7)

6.3 Color.- The color chip furnished by the procuring agency will be color chip No. X-24087, and may be obtained upon request to Commanding General, Aberdeen Proving Ground, ATTN: Coating and Chemical Laboratory, Aberdeen, Maryland 21005. (See 3.5)

6.4 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: T-23 W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.6.3.3)

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited to severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 A standard Microphone T-23( ) shall be loaned to the contractor for the duration of the contract. (See 3.3.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

6.8 Test chest.- Where specified in section 4, the contractor should use a test chest containing a piston-phone for activating the microphone. The chest volume and piston phone dimensions should be as follows:

Chest volume	4845 ± 3 cu. in.
Piston stroke	0.06 ± 0.002 in.
Piston diameter	0.7492 maximum, 0.7487 minimum

The dimensions of the test chest should be such that the approved standard Microphone T-23( ) will give 0.5 volt response when activated at the peak response frequency per Drawing SC-A-13474.

6.8.1 The piston phone shall be driven by a variable speed motor through a belt drive system.

6.8.2 The chest shall be so constructed that the leakage rate shall be such that the initial pressure applied shall not decrease to one-half of its original value in less than 10 seconds.

6.9 Audit.- An audit (see 4.5.1) is an assesment of the effectiveness of an act, a performance, or a process.

Custodian:
Army-EL

Preparing activity:
Army - EL
Project No. 5895-A179

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:06Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-12621C(SL) MICROPHONE, RIBBON (Microphone, T-23())

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-PM 1068-68

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:06Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL?P-ED
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:06Z
Check the source to verify that this is the current version before use.